Recap of (some) pre-requesite for this course

This short lecture summarizes some of the key pre-requesite for this course. It covers the following topics:

- expectations
- unbiasedness
- consistency / convergence in probability
- law of large numbers
- convergence in distribution
- central limit theorem

It is not intended to be exhaustive but it provides a recap of elements that are key to understand the material of this course. We will provide other reminders during the course and, of course, please ask questions during the lectures or tutorials when you do not understand a step.
As much as possible we try to provide intuitions and, therefore, avoid formal definitions/expositions when not necessary. For these missing formal definitions/expositions, please refer to your previous courses.

Expectations

Expectations for discrete variables

We start with discrete variables taking an *infinite* number of possible values.
Let X be a discrete random with support $\{\tau_j\}_{j=1}^{\infty}$. Let $\pi_j = \mathbb{P}\left[X = \tau_j\right]$. The expectation of X is defined as

$$\mathbb{E}[X] \equiv \sum_{j=1}^{\infty} \pi_j \tau_j$$

if the series $\{\sum_{j=1}^{n} \pi_j \tau_j\}_{n=1}^{\infty}$ is **convergent**.
Importantly, $\lim_{n\to\infty} \sum_{j=1}^{n} \pi_j \tau_j$ might not exist, in which case the expectation does not exist. It is also possible that $\lim_{n\to\infty} \sum_{j=1}^{n} \pi_j \tau_j = +\infty \text{ or } -\infty$ in which case the expectation is not finite.

Most properties in this course will require the existence of finite expectations for some random variables. As a result, these properties will not apply to random variables with non-finite expectation or for which the expectation does not exist.
Later in this recap, we will show examples of random variables with non-finite or non-existent expectations, and we illustrate that the law of large numbers does not apply for these variables.
Notice that for a discrete variable X taking $N < \infty$ values, the expectation is $\mathbb{E}[X] \equiv \sum_{j=1}^{N} \pi_j \tau_j$.

Expectation of a function

Consider a function $g(X)$ and the same assumptions as before. The expectation of $g(X)$ is defined as

$$\mathbb{E}[g(X)] \equiv \sum_{j=1}^{\infty} \pi_j g(\tau_j)$$

if the series $\{\sum_{j=1}^{n} \pi_j g(\tau_j)\}_{n=1}^{\infty}$ is **convergent**.
Note that even if $\mathbb{E}[X]$ exists and is finite, nothing ensures that $\mathbb{E}[g(X)]$ exists and is finite (and vice-versa).

Expectation of a continuous random variable

We can extend these definitions to the case of continuous random variables.
Let X be a continuously distributed random variable with density $f(X)$. The expectation of X is defined as

$$\mathbb{E}[X] \equiv \int_{-\infty}^{+\infty} x f(x) dx$$

if the integral is **convergent**.
Given what we have seen, the definition of $\mathbb{E}[g(X)]$ for continuous random variables should be straightforward.

Conditional expectation and related properties

We will not provide a formal definition of a conditional expectation, but provide intuitions for it.
Consider the expectation of the variable X representing here personal income. Let Z be a variable equal to 1 if a person is a woman and 0 if a person is a man (for simplicity of the exposition, we restrict our discussion to a situation in which gender is assumed to be binary).
If personal income is on average lower for women than for men, we would have:

$$\mathbb{E}(X \mid Z = 1) < \mathbb{E}(X \mid Z = 0)$$

where $\mathbb{E}(X \mid Z = 1)$ denotes expected income for women.

Denote the probability to be a woman by $\pi_{woman} = \mathbb{P}(Z = 1)$ and the one to be a man is $\pi_{man} = 1 - \mathbb{P}(Z = 1) = \mathbb{P}(Z = 0)$. Then, expected income in the population is:

$$\begin{aligned}\mathbb{E}(X) &= \pi_{woman} \times \mathbb{E}(X \mid Z = 1) + \pi_{man} \times \mathbb{E}(X \mid Z = 0) \\ &= \mathbb{P}(Z = 1) \times \mathbb{E}(X \mid Z = 1) + \mathbb{P}(Z = 0) \times \mathbb{E}(X \mid Z = 0)\end{aligned}$$

If you replace expectation by average, you can think of it as the fact that average income of a population is the weighted average of the income of women and men.

The previous formula is a special case of the following version of the law of iterated expectations:

$$\mathbb{E}(X) = \mathbb{E}[\mathbb{E}[X \mid Z]]$$

To see this, first notice that $\mathbb{E}[X \mid Z]$ is a *function* returning expected income for people with characteristics Z, so it is just a (non-random) function of Z which we can denote by $g(Z)$. The above formula can then be rewritten as:

$$\mathbb{E}(X) = \mathbb{E}[g(Z)]$$

Applying it to our example in which X is personal income and $Z = 0,\ 1$, and using the definition of the expectation of a function, we can rewrite the above expression as:

$$\mathbb{E}(X) = \pi_{woman} \times g(1) + \pi_{man} \times g(0)$$

or, using the fact that $g(1) = \mathbb{E}[X \mid Z = 1]$,

$$\mathbb{E}(X) = \pi_{woman} \times \mathbb{E}(X \mid Z = 1) + \pi_{man} \times \mathbb{E}(X \mid Z = 0)$$

It is very important that you know and that you understand the formula

$$\mathbb{E}(X) = \mathbb{E}[\mathbb{E}[X \mid Z]]$$

and that you remember that $\mathbb{E}[X \mid Z]$ is a function of Z.

X and Z could be vectors or matrices. For instance, we could extend our example by considering

$$X = \begin{pmatrix} \text{personal income} \\ \text{hours worked} \end{pmatrix}$$

$$Z = \begin{pmatrix} \text{gender} \\ \text{university degree} \end{pmatrix}$$

The first element of X collects personal income and the second collects hours worked, while Z collects information on gender (1 if woman, 0 if man) and whether a person has a university degree (1 if has a university degree, 0 if not). So,

$$\mathbb{E}\left(X \mid Z = \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right)$$

is a (2×1) matrix (as it is of the same dimension as X) containing expected income and hours worked for a woman with a university degree.

And expected income in the population is:

$$
\begin{aligned}
\mathbb{E}(X) &= \mathbb{E}[\mathbb{E}[X \mid Z]] \\
&= \mathbb{P}\left(Z = \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) \times \mathbb{E}\left(X \mid Z = \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) \\
&+ \mathbb{P}\left(Z = \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \times \mathbb{E}\left(X \mid Z = \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \\
&+ \mathbb{P}\left(Z = \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) \times \mathbb{E}\left(X \mid Z = \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) \\
&+ \mathbb{P}\left(Z = \begin{pmatrix} 0 \\ 0 \end{pmatrix}\right) \times \mathbb{E}\left(X \mid Z = \begin{pmatrix} 0 \\ 0 \end{pmatrix}\right)
\end{aligned}
$$

A more general version of the law of iterated expectations is

$$\mathbb{E}[X \mid Z_1] = \mathbb{E}\Big[\mathbb{E}[X \mid Z_1, Z_2] \mid Z_1\Big]$$

Suppose X is income, Z_1 is gender, and Z_2 is university degree. Then the above formula states that expected (think of average) income for a given gender Z_1 is a weighted average of $\mathbb{E}[X \mid Z_1, Z_2]$, where the latter represents expected income conditional on being of gender Z_1 and university degree Z_2.

So if $Z_1 = 1$, the above implies:

$$\begin{aligned} \mathbb{E}[X \mid Z_1 = 1] &= \mathbb{P}\left[Z_2 = 0 \mid Z_1 = 1\right] \times \mathbb{E}[X \mid Z_1 = 1, Z_2 = 0] && (1) \\ &+ \mathbb{P}\left[Z_2 = 1 \mid Z_1 = 1\right] \times \mathbb{E}[X \mid Z_1 = 1, Z_2 = 1] && (2) \end{aligned}$$

Finally, we have

$$\mathbb{E}\left[g\left(Z_1\right) f\left(Z_2\right) \mid Z_1\right] = g\left(Z_1\right) \mathbb{E}\left[f\left(Z_2\right) \mid Z_1\right]$$

and

$$\mathbb{E}\left[g\left(Z_1\right) \mid Z_1\right] = g\left(Z_1\right)$$

Asymptotic theory

In this part, we will:

- illustrate the difference between unbiasedness and consistency
- discuss the law of large numbers needed for this course
- discuss the notion of convergence in distribution
- discuss the central limit theorem

Illustration of unbiasedness

- consider the simple i.i.d. process

$$X_i \sim \mathcal{N}(\mu, \sigma^2)$$

- and consider drawing n^s random samples from this process, each of size $n = 10$.
- we are going to study what happens when the number of samples n^s increases while the size of a sample n remains equal to 10

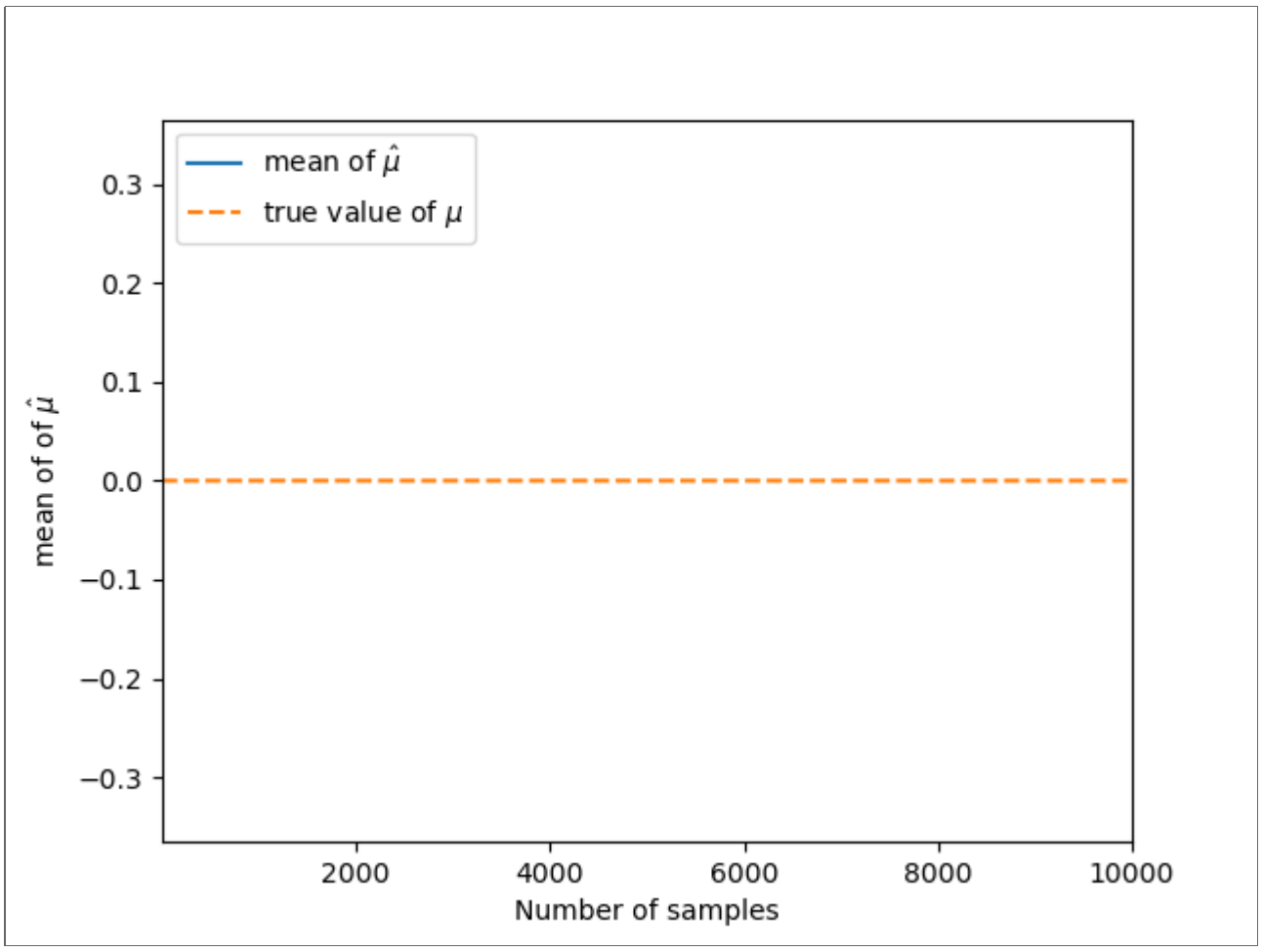
- more specifically, for each sample $i = 1, \ldots, n^s$ of size $n = 10$, we compute the sample mean $\hat{\mu}_i$, and then we compute the average of these (which is a random variable):

$$\overline{\hat{\mu}}_{n^s} = \frac{1}{n^s} \sum_{i=1}^{n^s} \hat{\mu}_i$$

and we observe the behaviour of $\overline{\hat{\mu}}_{n^s}$ as n^s increases.

- in doing so, we are going to illustrate that the sample mean $\hat{\mu}$ is an unbiased estimator of μ
- we start with a case where $\mu = 0$ and $\sigma^2 = 5$.

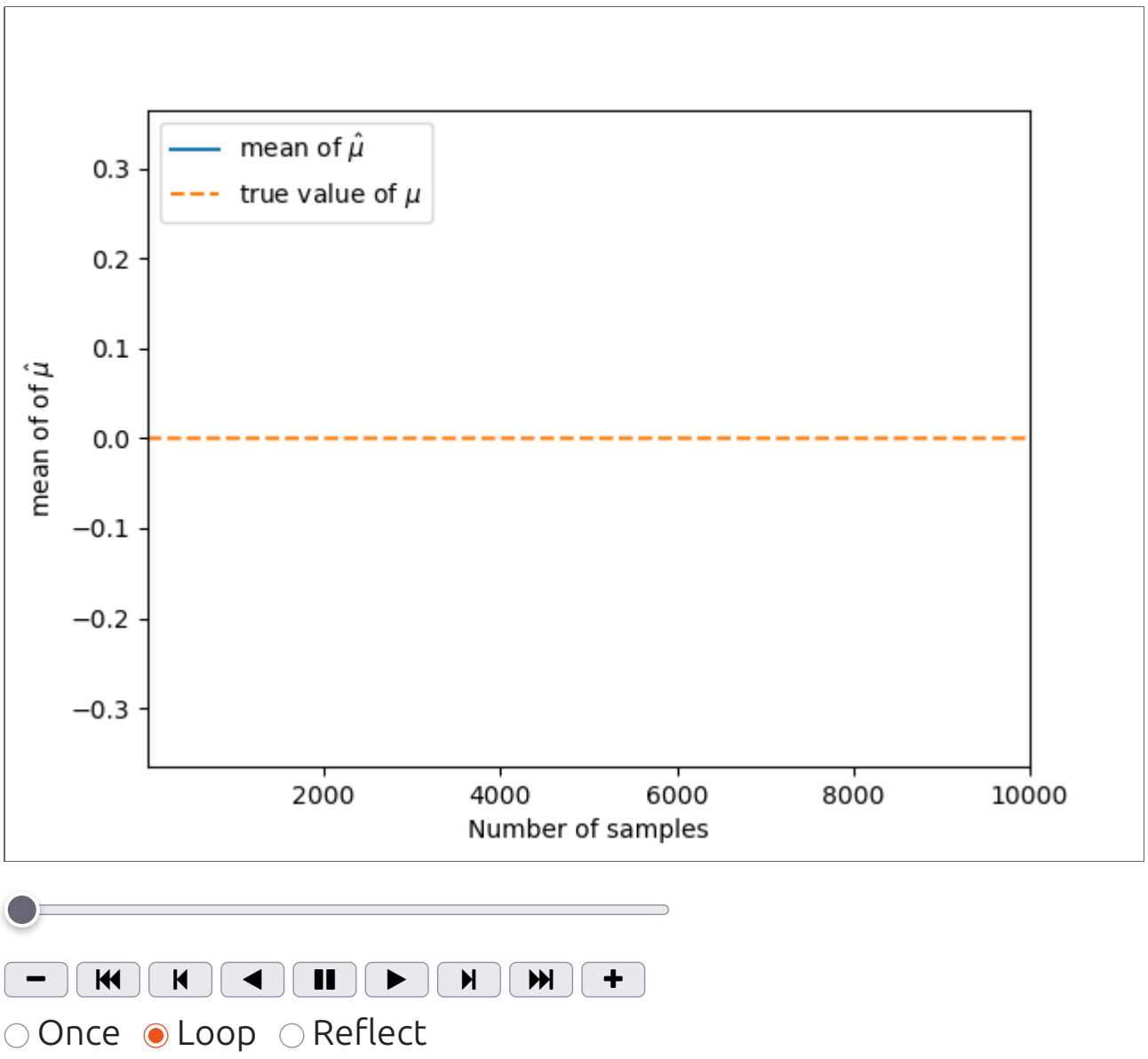
Out[3]:



Once Loop Reflect

If σ is lower or n larger, convergence is in general faster. Here is a case with $\sigma = 1$.

Out[5]:



Once Loop Reflect

- We have illustrated the fact that the sample mean is unbiased for μ here
- That is that, **for n fixed**:

$$\mathbb{E}(\hat{\mu}) = \mu$$

- The key intuition to get is that (loosely speaking), if an estimator is unbiased for some parameter, we can average the estimates from many small (independent) samples and we will get something close to the true value of this parameter.
- In the simple example here (using the sample mean as an estimator for the expectation), the sample mean is both unbiased and consistent (the next concept we discuss). However, later in this course, we will see that there are e.g. estimators that are consistent but not unbiased.

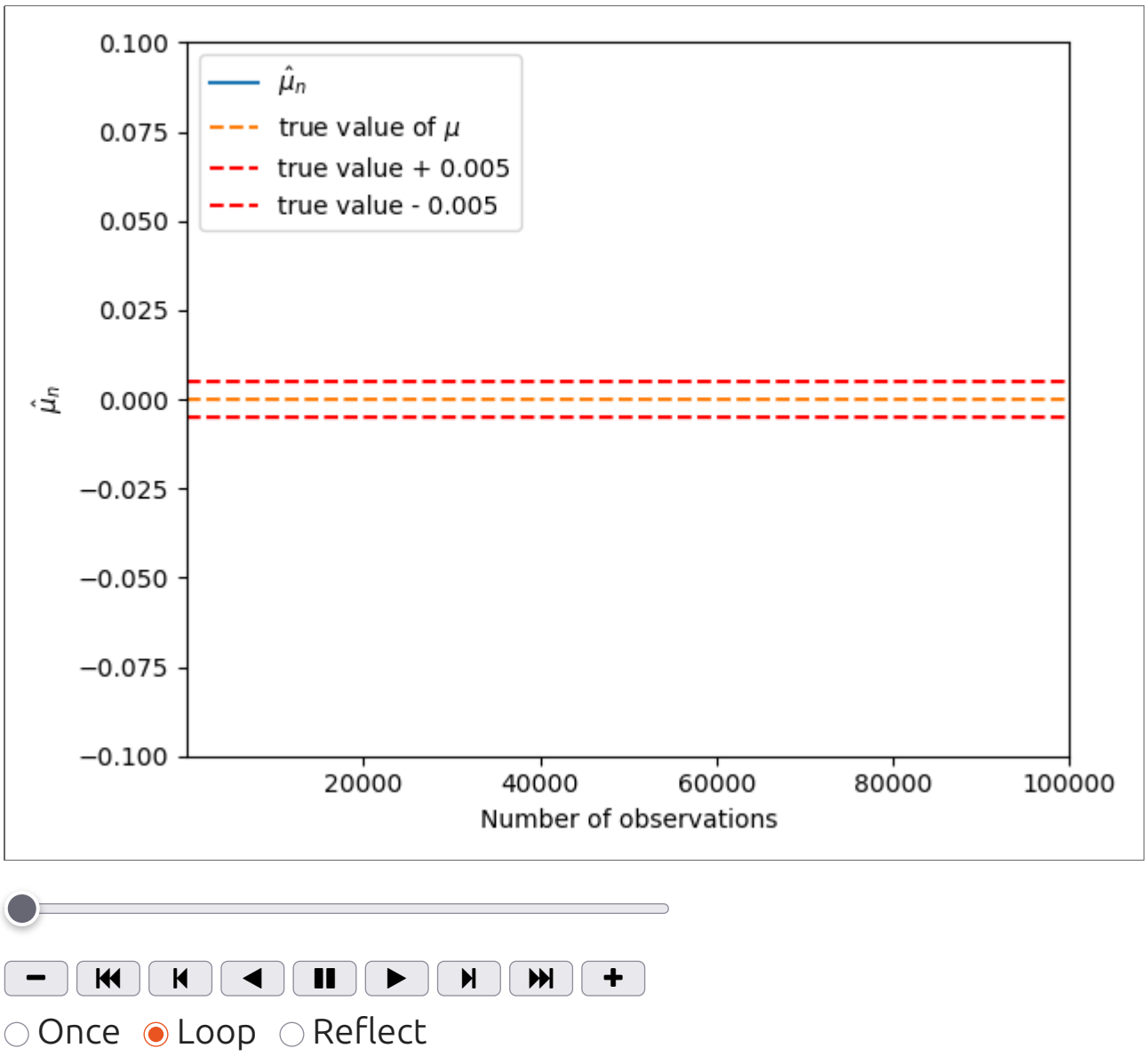
Illustration of consistency

- consider again the simple i.i.d. process

$$X_i \sim \mathcal{N}(\mu, \sigma^2)$$

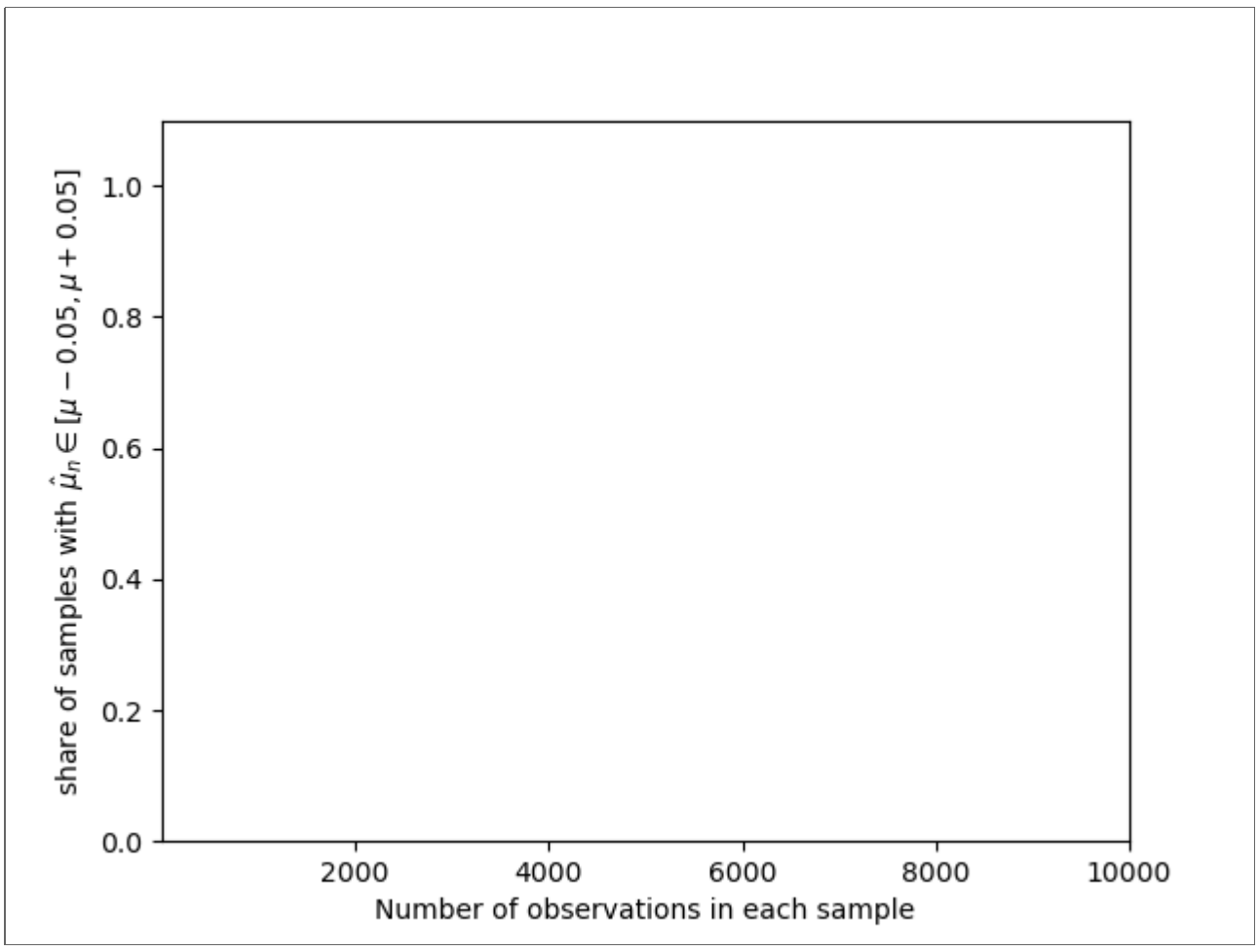
- and consider computing the average $\hat{\mu}_n$ of a large sample n made of draws from this process. What happens when n increases? (we index $\hat{\mu}$ by n to reflect the fact that this is a random sequence that depends on n)
- we consider the case where $\mu = 0$ and $\sigma^2 = 5$, and study at what happens when $n \to 100,000$.

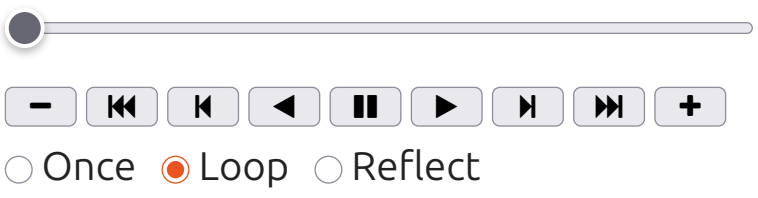
Out[7]:


0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
$\hat{\mu}_n$
$\hat{\mu}_n$
true value of μ
true value + 0.005
true value - 0.005
20000
40000
60000
80000
100000
Number of observations
Once
Loop
Reflect

- now we are going to repeat this (increase the size of the sample) for a large number of samples
- specifically, we take a rather large number of samples n^s (this number will remain fixed) each with n observations
- for each of these samples, we are going to compute $\hat{\mu}_n$
- and we are going to compute the share of samples for which $\hat{\mu}_n \in [\mu - 0.05, \mu + 0.05]$ as n increases

Out[9]:





Once Loop Reflect

This exercise illustrates the concept of consistency. An estimator is consistent for a parameter if it converges in probability to this parameter.
We have seen that, as the size of our samples increases, it becomes unlikely that the sample mean of a sample falls outside of some interval around the expectation of our process.
The key intuition to get is that (loosely speaking), if an estimator is consistent for a parameter, one sufficiently large sample is "enough" to get an estimate that is close to the true value of this parameter.

Convergence in probability: formal definition

Formally, a sequence of random scalars $\{z_n\}$ *converges in probability* to a constant (nonrandom) α if, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} Pr(|z_n - \alpha| > \varepsilon) = 0$$

We write: "$\underset{n\to\infty}{plim}\, z_n = \alpha$" or as "$z_n \underset{p}{\to} \alpha$" or as "$z_n - \alpha \underset{p}{\to} 0$".

- The definition also applies to vectors: in that case, each element of the random vector must converge in probability to the corresponding element of the limit (non-radom) vector
- Convergence in probability can also be to a **random** vector (or scalar) $\mathbf{z}$ which occurs when $\tilde{\mathbf{z}}_n \equiv \mathbf{z}_n - \mathbf{z} \underset{p}{\to} \mathbf{0}$ which is also written $\mathbf{z}_n \underset{p}{\to} \mathbf{z}$

In our example above, our random sequence is the one for the sample mean of a sample of size n $\{\hat{\mu}_n\}$, and we have illustrated numerically the fact that:

$$\hat{\mu}_n \underset{p}{\rightarrow} \mu = 0$$

i.e. that for any $\varepsilon > 0$:

$$\lim_{n \to \infty} \text{Prob}(\mid \hat{\mu}_n - \mu \mid > \varepsilon) = 0$$

A law of large numbers

The fact that $\hat{\mu}_n \underset{p}{\rightarrow} \mu$ in the previous example is an illustration of the following law of large numbers, which is the only one you need to remember for this course.

Kolmogorov's Second Strong Law of Large Numbers

Let $\{\mathbf{z}_i\}$ be i.i.d. with $\mathbb{E}(\mathbf{z}_i) = \boldsymbol{\mu}$ where $\boldsymbol{\mu}$ exists and is finite, then:

$$\bar{\mathbf{z}}_n \equiv \frac{1}{n}\sum_{i=1}^{n} \mathbf{z}_i \underset{p}{\rightarrow} \boldsymbol{\mu}$$

(Note: this law is normally written with almost sure convergence (see Hayashi p95), but almost sure convergence implies convergence in probability which is all we need here. That is why we state it in this way (in its weak form).)

In our example, given that the process was i.i.d. and that X was normally distributed with expectation $\mu = 0$, the above theorem applied. Therefore, the averages of i.i.d. draws from this process were converging in probability to the true expectation.
Importantly, we need 2 assumptions for this theorem to be applicable: observations must be i.i.d. and the expectation needs to exist and be finite. Make sure to check them and state them explicitly when you use this theorem (failure to do so leads to fewer points at the exam).

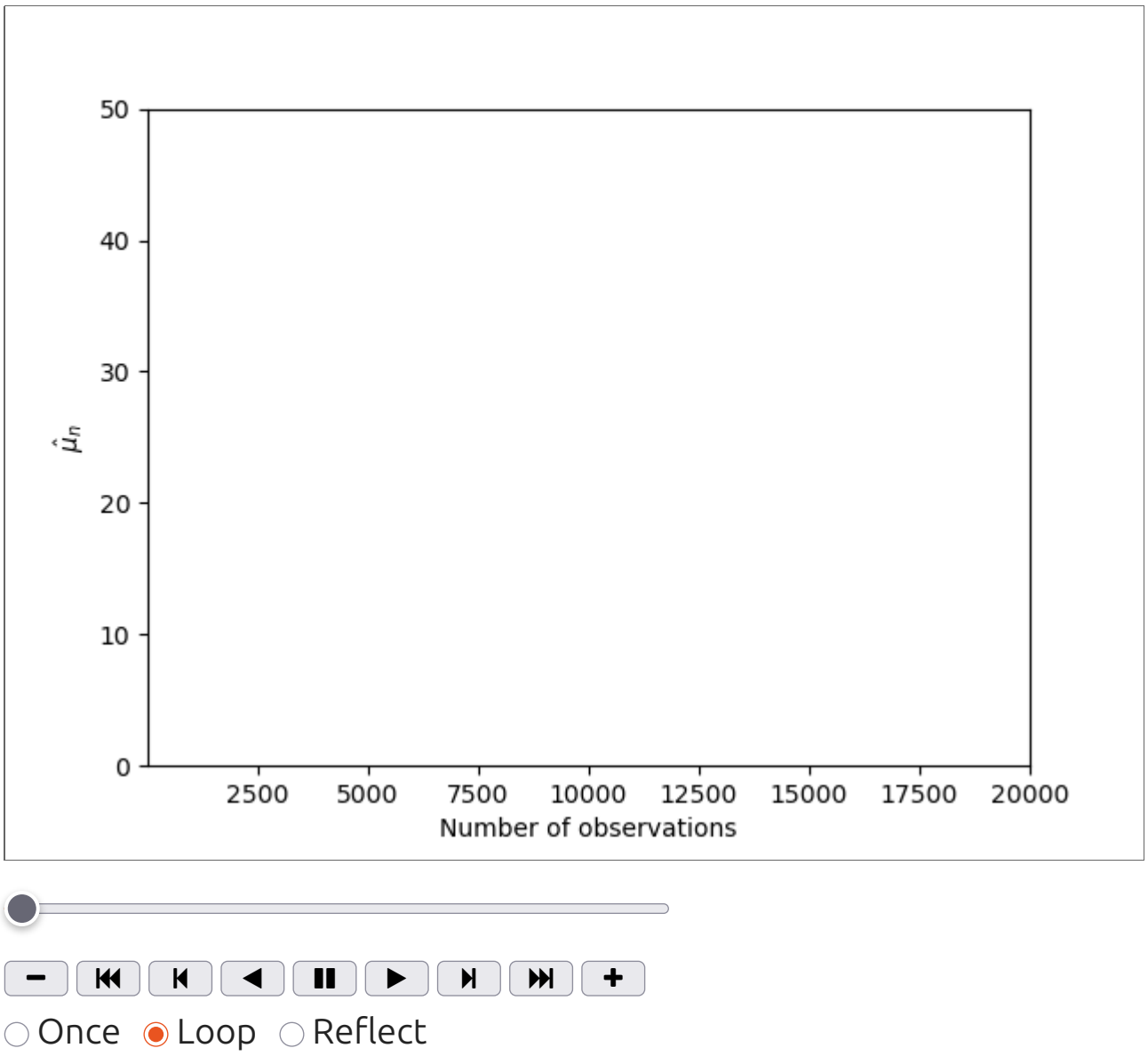
Failure of the law of large numbers when the expectation is infinite

Consider the following game (known as St. Petersburg's paradox). Flip a fair coin until a head appears. The payout X of the game is 2^k where k is the number of times the coin has been flipped before a head was reached.
The expectation of X is:

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \pi_k 2^k = \sum_{k=1}^{\infty} 2^{-k} 2^k = +\infty$$

Given that the expectation is infinite, the above LLN does not apply and, therefore, we should not expect that, repeating the game many times, the average of the payouts would converge to a finite value.

Out[12]:


50
40
30
20
10
0
$\hat{\mu}_n$
2500
5000
7500
10000
12500
15000
17500
20000
Number of observations
Once
Loop
Reflect

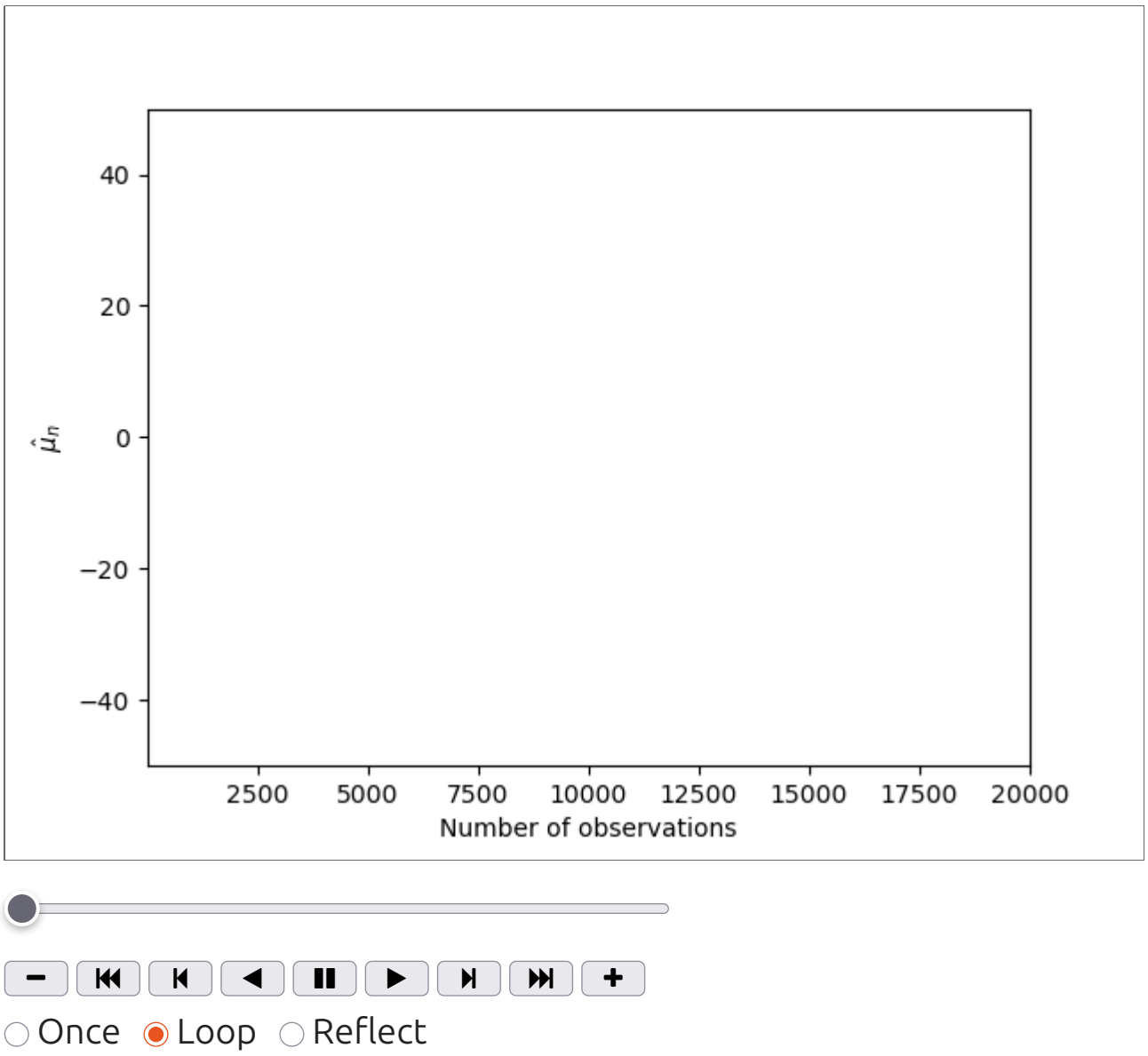
Failure of the law of large number when the expectation does not exist

Consider the following modified version of the St. Petersburg's paradox for which the expectation does not exist. Flip a fair coin until a head appears. Once a head appears, flip the coin one last time. If a head appears the payout X is 2^k, otherwise the payout is -2^k. k is here the number of times the coin has been flipped before the first head appeared.
The expectation of X is:

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \frac{2^{-k}}{2} \times 2^k + \sum_{k=1}^{\infty} \frac{2^{-k}}{2} \times (-2^k) = \sum_{k=1}^{\infty} \frac{1}{2} - \sum_{k=1}^{\infty} \frac{1}{2} = \infty - \infty$$

and it can be shown that the two limits do not cancel each other out (see video **here** for an explanation).
Given that the expectation does not exist, the above LLN does not apply and, therefore, we should not expect that, repeating the game many times, the average of the payouts would converge to a finite value.

Out[14]:


40
20
0
−20
−40
$\hat{\mu}_n$
2500
5000
7500
10000
12500
15000
17500
20000
Number of observations
Once
Loop
Reflect

Failure of i.i.d. assumption

Consider the following time series process (think here as n being a year):

$$X_n = \begin{cases} -1 \ \forall n \geq 1 & \text{with probability } 1/2 \\ 1 \ \forall n \geq 1 & \text{with probability } 1/2 \end{cases}$$

which is equivalent to have a coined being flipped at time 1 which determines the value of X_n for all future periods. The expectation of this process is 0. However, if we have access to an infinite time-series of this process, the sample mean will be either -1 or 1. So, obviously, the sample mean is an inconsitent estimator of the expectation in this case.

The law of large numbers fails here because the observations are not independent. Although this case is extreme, the failure of independence of observations is inherent to time-series econometrics (something you will see next block).

So, once again, if you use the LLN, make sure to check the two assumptions needed to apply it and make sure to write it in your exam.

Convergence in distribution

Let F_n be the cumulative distribution function (c.d.f.) of z_n.

- $\{z_n\}$ **converges in distribution** to a random scalar z if

$$\lim_{n\to\infty} F_n(z) = F(z)$$

for all points z where $F(\,.\,)$ is continuous.

- also applicable to vectors (important: convergence element by element not sufficient - see Hayashi p91; intuitive reason: need to account for the fact that the different elements might not be independent - for instance, if correlated)
- convergence in probability implies convergence in (degenerate) distribution

Convergence element by element is not sufficient

Suppose that:

$$\begin{pmatrix} x_n^1 \\ x_n^2 \end{pmatrix} \underset{d}{\rightarrow} \begin{pmatrix} x^1 \\ x^2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}\right)$$

and that there is

$$\begin{pmatrix} z^1 \\ z^2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{22} \end{pmatrix}\right)$$

then:

$$x_n^1 \underset{d}{\rightarrow} z^1$$

$$x_n^2 \underset{d}{\rightarrow} z^2$$

as both x_n^i and z_n^i follow $\mathcal{N}\left(\mu_i, \sigma_{ii}\right)$. Yet, $\begin{pmatrix} x_n^1 \\ x_n^2 \end{pmatrix}$ does not converge in distribution to $\begin{pmatrix} z^1 \\ z^2 \end{pmatrix}$.

Useful results

Preservation of convergence for continuous transformation (Lemma 2.3)
$\mathbf{a}()$ is a vector-valued **continuous** function that **does not depend on** n
.
Then
(a). $\mathbf{z}_n \underset{p}{\rightarrow} \alpha \Rightarrow \mathbf{a}\left(\mathbf{z_n}\right) \underset{p}{\rightarrow} \mathbf{a}\left(\alpha\right)$
(b). $\mathbf{z}_n \underset{d}{\rightarrow} \alpha \Rightarrow \mathbf{a}\left(\mathbf{z_n}\right) \underset{d}{\rightarrow} \mathbf{a}\left(\alpha\right)$
Examples of $\mathbf{a()}$: addition, multiplication, division, matrix inversion
(note: the lemma indeed easily generalizes to matrices as a matrix can always be recasted as a vector)

Lemma 2.4. (Parts (a) and (c) are called the Slutzky's theorem)

(a). $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x},\ \mathbf{y}_n \underset{p}{\rightarrow} \boldsymbol{\alpha} \Rightarrow \mathbf{x}_n + \mathbf{y}_n \underset{d}{\rightarrow} \mathbf{x} + \boldsymbol{\alpha}$

(b). $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x},\ \mathbf{y}_n \underset{p}{\rightarrow} \mathbf{0} \Rightarrow \mathbf{y}_n' \mathbf{x}_n \underset{p}{\rightarrow} 0$

(c). $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x},\ \mathbf{A}_n \underset{p}{\rightarrow} \mathbf{A} \Rightarrow \mathbf{A}_n \mathbf{x}_n \underset{d}{\rightarrow} \mathbf{A}\,\mathbf{x}$

(d). $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x},\ \mathbf{A}_n \underset{p}{\rightarrow} \mathbf{A} \Rightarrow \mathbf{x}_n' \mathbf{A}_n^{-1} \mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x}' \mathbf{A}^{-1} \mathbf{x}$

Application of: *(c)*. $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x}$, $\mathbf{A}_n \underset{p}{\rightarrow} \mathbf{A} \Rightarrow \mathbf{A}_n \, \mathbf{x}_n \underset{d}{\rightarrow} \mathbf{A} \, \mathbf{x}$

Show that:

$$t_k \equiv \frac{\sqrt{n}\left(b_k - \beta_k\right)}{\sqrt{\widehat{Avar}\left(b_k\right)}} \underset{d}{\longrightarrow} \mathcal{N}\left(0, 1\right)$$

if

$$\sqrt{n}\left(b_k - \overline{\beta}_k\right) \underset{d}{\longrightarrow} \mathcal{N}\left(0, Avar\left(b_k\right)\right)$$

$$\widehat{Avar}\left(b_k\right) \underset{p}{\longrightarrow} Avar\left(b_k\right)$$

A central limit theorem

Lindeberg-Levy CLT
Let $\{\mathbf{z}_i\}$ be i.i.d. with $E(\mathbf{z}_i) = \boldsymbol{\mu}$ and $Var(\mathbf{z}_i) = \boldsymbol{\Sigma}$. Then

$$\sqrt{n}(\bar{\mathbf{z}}_n - \boldsymbol{\mu}) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (\mathbf{z}_i - \boldsymbol{\mu}) \underset{d}{\rightarrow} N(\mathbf{0}, \boldsymbol{\Sigma})$$

The additional assumption needed here is the existence and finitness of the variance (which is a function of $\mathbf{z}_i$).

References

- Bruce E. Hansen, "Probability and Statistics for Economists", Princeton University Press
- Fumio Hayashi, "Econometrics", Princeton University Press